PYTHON BASICS

Complete Beginner Notes • Interview Revision

Simple explanations, code examples, and outputs

Study tip: Pratyek code adhi swatah type kar, run kar, ani output ka ala te swatah samjun sang. Fakt path karu nakos.

Prepared for: Python-Programming GitHub practice repository

1. Python mhanje kay?

Python hi beginner-friendly programming language aahe. Web development, automation, data analysis, AI/ML ani scripting sathi vapartat.

- Python code line-by-line execute hoto.
- Python case-sensitive aahe: `name` ani `Name` vegale variables aahet.
- Indentation mahatvache aahe. if/for/while/function chya block madhye spaces vapra.

```
print("Hello, Python!")
```

Output: Hello, Python!

2. print(), comments ani variables

print() — output dakhavto

```
print("Hello")
print(10)
print(2 + 3)
```

Output: Hello, 10, 5 (pratyek print navin line var).

Comments

```
# Ha comment aahe; Python execute karat nahi
print("Hi")
```

Variables

```
name = "Raj"
age = 23
marks = 9.23
is_student = True
print(name, age, marks, is_student)
```

Variable mhanje value store karanyasathi nav. Python madhye type lihinyachi garaj naste.

- Variable names letter kiwa underscore ne suru kara; number ne suru karu naka.
- Spaces nako: `student_name` barobar; `student name` chukiche.

3. Basic data types

```
age = 23             # int: whole number
price = 99.5         # float: decimal
name = "Raj"         # str: text
is_ready = True      # bool: True / False
items = [1, 2, 3]    # list
point = (4, 5)       # tuple
```

- **int:** 10, -4
- **float:** 3.14, 9.5
- **str:** 'Python', "Hello"
- **bool:** True kiwa False (capital letters lakshat theva)

```
x = 10
print(type(x))
```

Output: `<class 'int'>`

4. Input ani type conversion

input() ne user kadun value ghene

```
name = input("Enter name: ")
print("Hello", name)
```

Mahatvache: input() default ne string deto.

int(), float(), str()

```
a = int(input("Enter number: "))
b = int(input("Enter another number: "))
print(a + b)
```

User 5 ani 3 dila tar output 8. int() mule input numbers madhye convert hotat.

```
n = 5
print(str(n) + " years")
```

Output: 5 years

5. Operators

Arithmetic operators

```
print(5 + 2)   # 7
print(5 - 2)   # 3
print(5 * 2)   # 10
print(5 / 2)   # 2.5
print(5 // 2)  # 2: floor division
print(5 % 2)   # 1: remainder
print(5 ** 2)  # 25: power
```

Comparison operators

`==, !=, >, <, >=, <=` he comparison karun True kiwa False detat.

```
print(10 == 10)  # True
print(10 != 5)   # True
print(3 > 5)     # False
```

Assignment vs comparison

- = value assign karto: `x = 10`
- == don values equal aahet ka check karto: `x == 10`

Logical operators

```
age = 23
print(age > 18 and age < 30)  # True
print(age < 18 or age == 23)   # True
print(not True)                # False
```

and = donhi conditions True pahijet; or = kamit kami ek True; not = result ulta karto.

6. if, elif, else

```
marks = 72
if marks >= 75:
    print("Distinction")
elif marks >= 40:
    print("Pass")
else:
    print("Fail")
```

Output: Pass

- if: pahili condition check karto.

- elif: adhi condition False asel tar dusri condition check karto.
- else: varchya conditions False astil tar chalto.

Indentation barobar theva; if/elif/else nantar colon (:) lagto.

7. Loops: for, range, while

for loop

```
for i in range(1, 5):
    print(i)
```

Output: 1, 2, 3, 4. range(1, 5) madhye 5 include hot nahi.

range()

- `range(5)` → 0, 1, 2, 3, 4
- `range(1, 5)` → 1, 2, 3, 4
- `range(1, 10, 2)` → 1, 3, 5, 7, 9

while loop

```
i = 1
while i <= 3:
    print(i)
    i += 1
```

Output: 1, 2, 3. while madhye condition True asel toparyant loop chalto. Update nahi kela tar infinite loop hou shakto.

break ani continue

```
for i in range(1, 6):
    if i == 3:
        break
    print(i)
# Output: 1, 2
```

```
for i in range(1, 6):
    if i == 3:
        continue
    print(i)
# Output: 1, 2, 4, 5
```

- **break:** purna loop thambavto.
- **continue:** current iteration skip karun pudhe chalto.

8. Strings

```
text = "Python"
print(text[0])       # P
print(text[-1])      # n
print(text[0:3])     # Pyt
print(text.upper())  # PYTHON
print(text.lower())  # python
print(len(text))    # 6
```

- Index 0 pasun suru hoto.
- Negative index -1 mhanje shevatcha character.
- Slicing `text[start:end]` madhye start include hoto, end include hot nahi.

```
name = "Raj"
print("Hello " + name)  # Hello Raj
```

String madhil characters badalta yet nahit (strings immutable aahet).

9. Lists

```
numbers = [10, 20, 30]
print(numbers[0])    # 10
print(numbers[-1])   # 30
numbers.append(40)
print(numbers)       # [10, 20, 30, 40]
numbers[1] = 50
print(numbers)       # [10, 50, 30, 40]
print(len(numbers))  # 4
```

- List square brackets `[]` vaparte.
- List mutable aahe: elements add, remove, update karta yetat.
- `append(value)` list chya shevti value add karto.
- `len(list)` elements chi sankhya deto.

List slicing

```
a = [1, 2, 3, 4, 5]
print(a[1:4])  # [2, 3, 4]
```

10. Tuples

```
point = (10, 20, 30)
print(point[0])  # 10
```

- Tuple round brackets `()` vaparte.
- Tuple immutable aahe: tuple madhil element direct change karta yet nahi.

Interview answer: List is mutable and uses square brackets. Tuple is immutable and uses parentheses.

11. Functions

```
def add(a, b):
    return a + b

print(add(5, 3))  # 8
```

- `def` function tayar karto.
- Parameters: function definition madhil inputs, udaharanarth a ani b.
- Arguments: function call kartana dilelya values, udaharanarth 5 ani 3.
- `return` result parat deto.

```
def greet(name):
    print("Hello", name)

greet("Raj")
# Output: Hello Raj
```

print() output dakhavto; return value function baher vaparta yete.

12. Local ani global variables

```
x = 10  # global variable

def test():
    x = 20  # local variable
    print(x)

test()     # 20
print(x)   # 10
```

Function madhil x ha local aahe; bahercha x global aahe. Same nav asle tari he don vegale variables asu shaktat.

13. Common beginner mistakes

- == ani = confuse karu naka.
- range(1, 5) madhye 5 yet nahi.
- List indexing 0 pasun suru hote.
- input() string deto; calculation sathi garaj asel tar int() / float() kara.
- if, for, while, def nantar colon (:) dya.
- Block madhye consistent 4 spaces indentation vapra.
- Infinite while loop talnyasathi variable update kara.
- Variable name case-sensitive aahe: Age ani age vegale.

14. Quick interview revision

Q. List ani tuple madhye farak?
List mutable aahe; tuple immutable aahe.

Q. break ani continue?
break loop purna thambavto; continue current iteration skip karto.

Q. /, //, % madhye farak?
/ division; // floor division; % remainder.

Q. len() kay karto?
String madhil characters kiwa list madhil elements mojto.

Q. append() kay karto?
List chya shevti element add karto.

Q. return kay karto?
Function madhun value parat deto.

Q. = ani ==?
= assignment; == comparison.

Q. Local variable?
Function chya aat define keleli variable.

Q. range(1, 4) values?
1, 2, 3.

Q. String upper/lower?
upper() uppercase; lower() lowercase.

15. Practice programs

He programs notes na baghta lihinyacha prayatna kara:

- 1. Don numbers input gheun addition print kara.
- 2. Number even aahe ki odd te check kara.
- 3. Number positive, negative ki zero te check kara.
- 4. 1 te 10 numbers print kara.
- 5. 1 te n paryant sum kadha.
- 6. 1 te 5 numbers print kara, pan 3 skip kara (continue).
- 7. String reverse kara.
- 8. String madhil characters chi sankhya mojha.

- 9. List madhye 4 values add karun print kara.
- 10. Function tayar kara jo don numbers chi sum return karel.
- 11. Number cha multiplication table print kara.
- 12. Number prime aahe ka check kara.

16. GitHub upload

Suggested repository: **Python-Programming**

PDF file name: `Python_Basics_Notes_20261012.pdf`

Suggested folder layout:

```
Python-Programming/
■■■ 01_Basics/
■■■ 02_If_Else/
■■■ 03_Loops/
■■■ 04_Strings/
■■■ 05_Lists_Tuples/
■■■ 06_Functions/
■■■ 07_Number_Problems/
■■■ 08_Patterns/
■■■ 09_Coding_Problems/
■■■ Python_Basics_Notes_20261012.pdf
■■■ README.md
```

GitHub var repository open kara → Add file → Upload files → PDF select kara → Commit changes. PDF sobat programs che .py files topic-wise upload kara.

Daily method: 20–30 minutes notes read kara, code swatah type/run kara, ani pratyek topic madhun 2–3 programs solve kara.